

Performance Evaluation of Pilot Scale Dual Column VPSA System for Post Combustion CO₂ Capture by Zeolite 13X Beads

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01 INTRODUCTION

Adsorption techniques including PSA, TSA, ESA, VPSA have gained considerable attention for post combustion CO₂ capture. VPSA offers low regeneration energy requirement & favourable cyclic performance VPSA emerged as promising alternative for CO₂ capture.

02 OBJECTIVES

To experimentally investigate pilot-scale post-combustion CO₂ capture from a simulated industrial flue gas using a dual-column Vacuum Pressure Swing Adsorption (VPSA) system packed with commercial Zeolite 13X beads.

03 METHODOLOGY

- ✓ The VPSA system consisted of two adsorption columns (60 kgs each) operated alternately under adsorption and desorption conditions.
- ✓ An optimized flue gas mixture containing air and CO₂ in an 80:20 ratio at 125 SLPM was supplied.

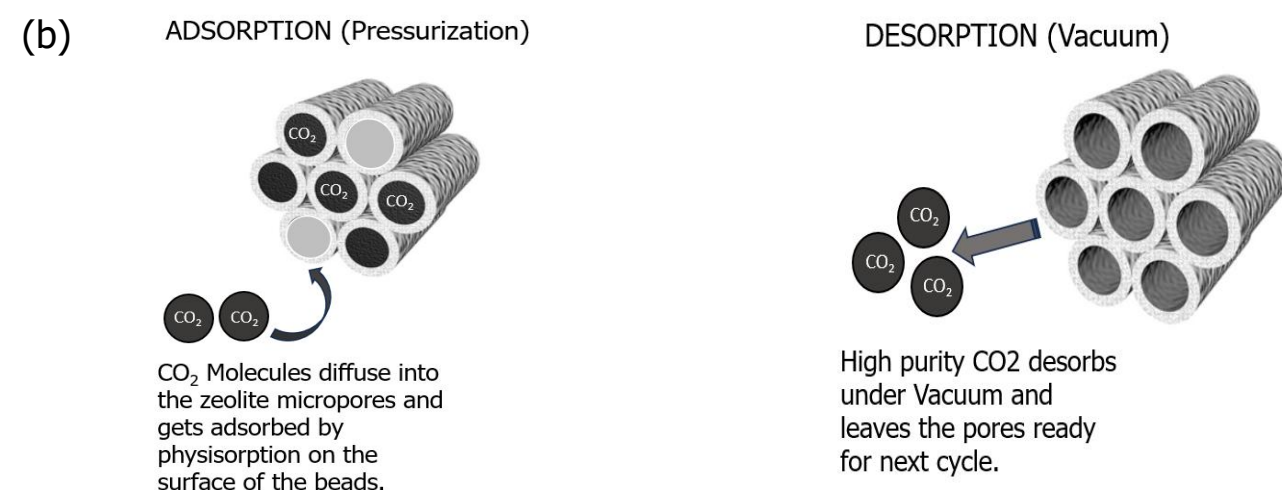
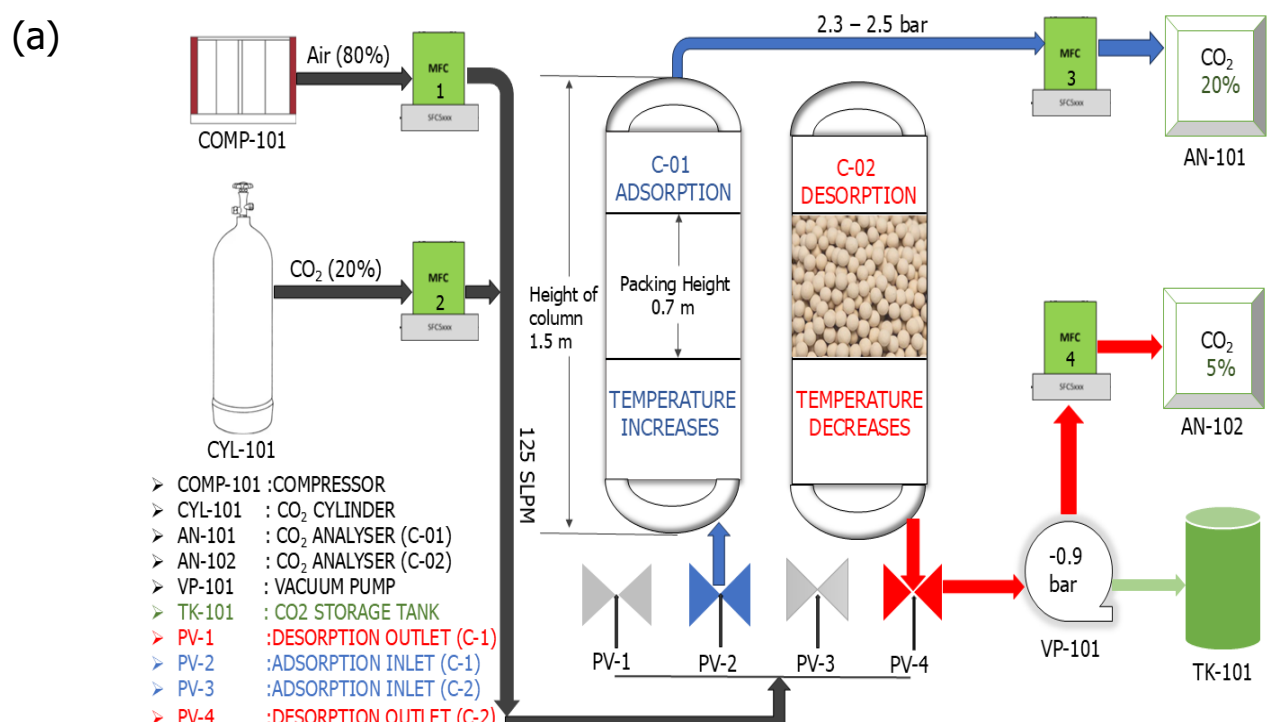


Figure 1. (a) Schematic representation (b) Separation Mechanism of VPSA Dual System

04 MATERIAL CHARACTERIZATION

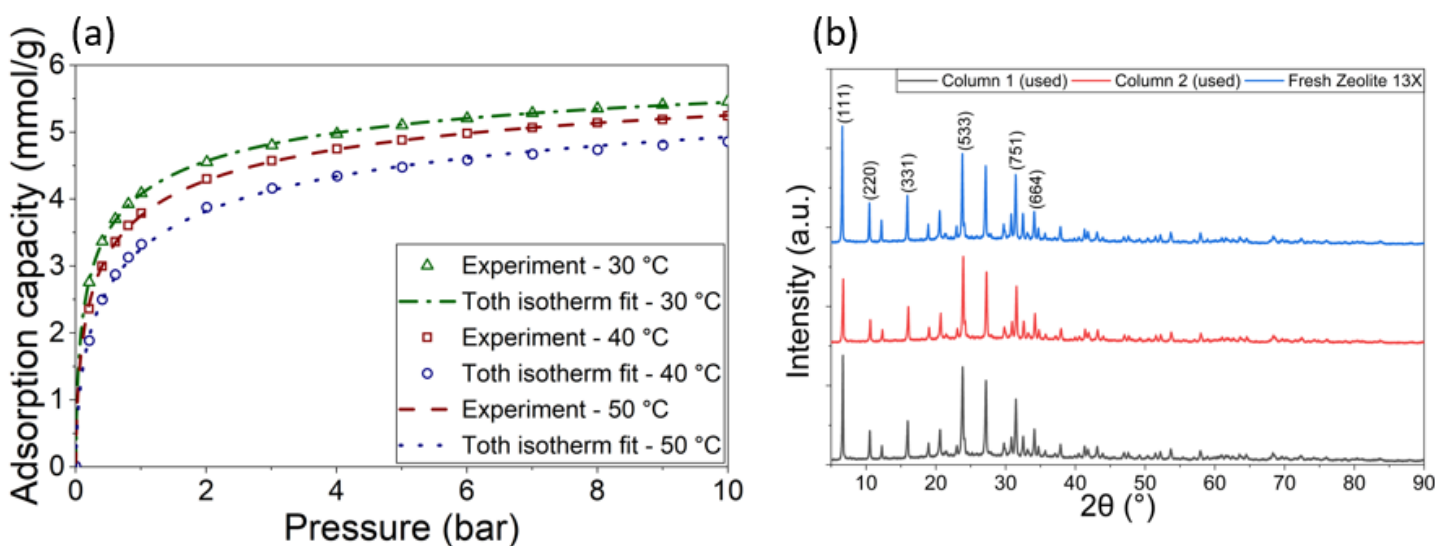


Figure 2. (a) High pressure adsorption (b) PXRD characterization of both columns with fresh zeolite

05 TEMPERATURE & CO₂ OUTLET CONCENTRATION

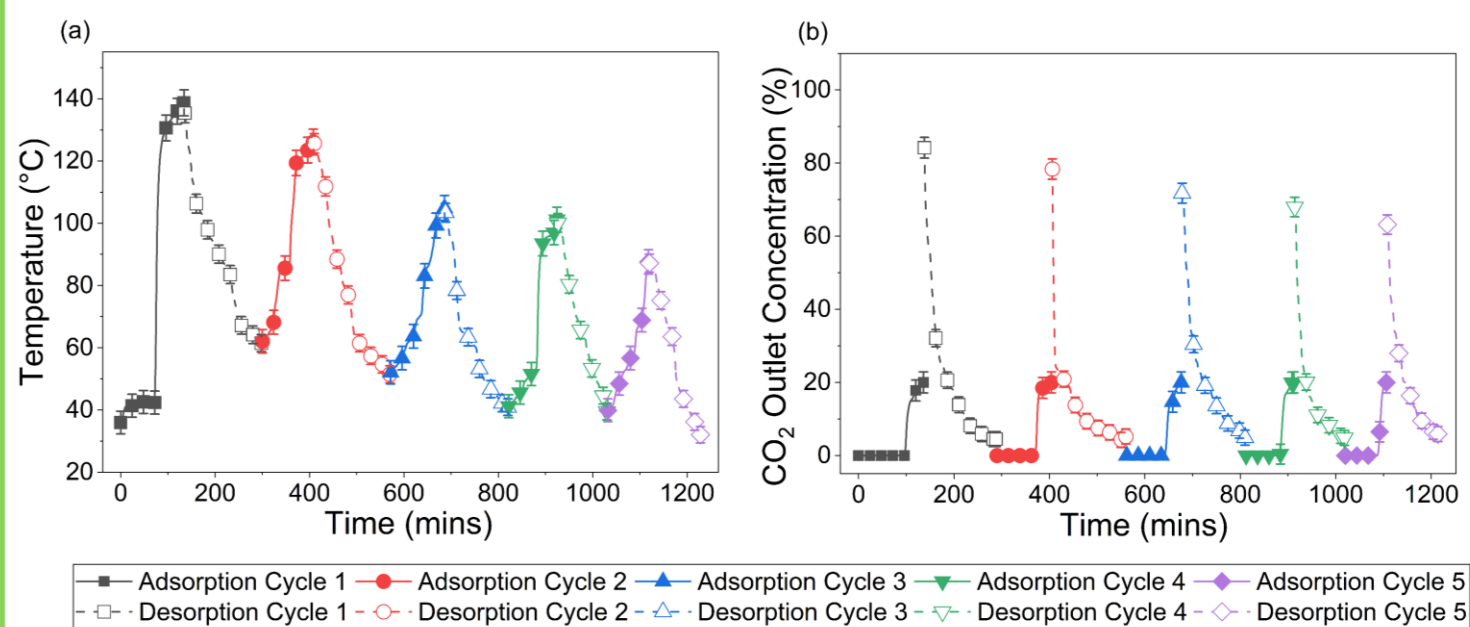


Figure 4. (a) Temperature Profile Analysis (b) CO₂ Outlet Concentration for Column-01

06 PRESSURE DROP & ADSORPTION CAPACITY

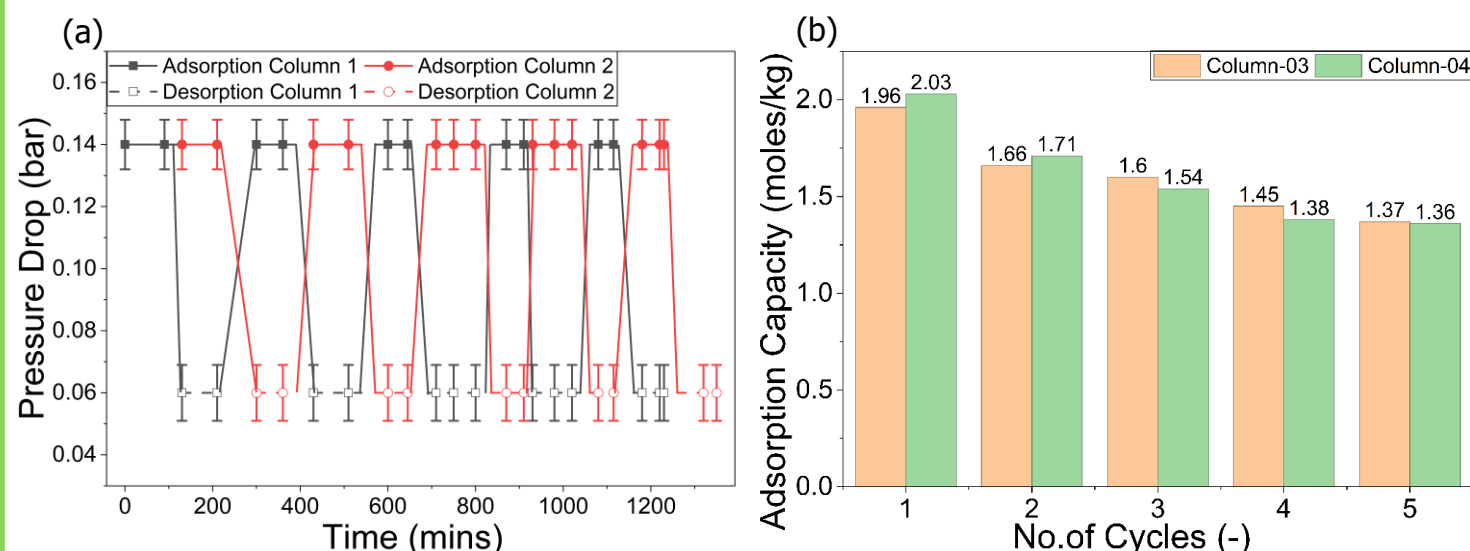


Figure 5. (a) Pressure drop (b) Adsorption capacities of both over five VPSA cycles

- ✓ Maximum CO₂ adsorption capacities of 1.96 & 2.03 moles/kg were obtained respectively

Table 1. Comparison of Mass of CO₂ Captured (Kg) for dual column system

No. of Cycle	Mass of CO ₂ captured (Kg)	
	Column-01	Column-02
1	5.19	5.38
2	4.39	4.54
3	4.22	4.06
4	3.83	3.65
5	3.62	3.6

- ✓ The system achieved Mass of CO₂ capture up to 5.19 & 5.38 kg/cycle for Column 1 and Column 2 respectively

07 CONCLUSIONS

- ✓ Stable cyclic operation confirms the feasibility of pilot scale VPSA for for Continuous CO₂ capture
- ✓ Commercial Zeolite 13X exhibited effective performance for industrial carbon capture applications

08 FUTURE WORK

- ✓ Development of a one-dimensional MATLAB model to predict pilot scale VPSA performance over 100 cycles for performance prediction and process optimization

09 ACKNOWLEDGEMENTS

